Electrical Topology Diagram (Implementation v6)

As-of date: `2026-07-19`

Purpose: provide a complete, install-level electrical topology for the current build scope, including all major electrical components, fuse IDs, fuse housings, planned wire gauges, and estimated one-way run lengths for procurement planning.

Related docs:

- Canonical electrical/system baseline: `docs/core/SYSTEMS.md`
- Detailed fuse matrix: `docs/implementation/ELECTRICAL_fuse_schedule.md`
- Battery and trunk recalculation record: `docs/studies/ELECTRICAL_battery_fuse_wire_recalc_2026-02-18.md`
- Decisions and unresolved items: `docs/core/TRACKING.md`
- Procurement source of truth: `bom/bom_estimated_items.csv`

Sweep Outcomes Included In This Revision

- Keeps alternator charging architecture on the dedicated `48V` secondary alternator path (`Mechman + WS500 + APM-48` baseline); obsolete pre-Mechman charger paths are removed from primary topology.
- Keeps Lynx Slot 3 branch to alternator input with `F-04 200A/80V` MEGA.
- Removes obsolete pre-Mechman engine-bay fuse/conductor placeholders from active architecture.
- Clarifies the confirmed `PH-VAN` harness as one `15A` Class-CC fused combined power/positive-sense red lead (`F-12/F-13-PHVAN`) and locks the unfused purple/grey pair to the Wakespeed shunt after the SmartShunt in the common battery-negative path.
- Adds Ford `Upfitter #3 -> F-15 -> WS500 brown ignition` manual alternator-control path.
- Adds the detailed Mechman/WS500/APM-48 install guide as the shop reference for staged installation, first-run checks, and load-dump/shutdown handling.
- Defines APM-48 as a parallel surge clamp at the alternator rather than a series charge-current device.
- Added explicit fuse-holder/housing definitions for every fuse family (`Class T` , Lynx `MEGA` , inline `MIDI/ANL/AMI` , PV `gPV` , and `ATO/ATC`).
- Added conductor schedule across `48V` , `12V` , PV, and AC segments with explicit assumptions.
- Updated 12V topology to a shared 12V junction fed by an Orion-Tr Smart `48/12-30` charger and a `12V 100Ah` buffer battery branch, with `F-11` source fuse plus `SW-12V-BATT` manual isolation.
- Added a full-circuit estimated run-length validation pass (`C-01` through `C-40`) and purchase-ready wire rollup totals.

Current Commissioning Snapshot (`2026-08-02`)

- `48V` bus live-tested at `55.5V` throughout the system, including at the MultiPlus.
- MultiPlus-II inverter mode tested with inverter light on, slight normal hum, and no reported errors.
- SmartShunt and Orion-Tr Smart connected in VictronConnect.
- Cerbo GX access point/remote-console workflow active; Cerbo is powered from a small inline fused `48V` feed and connected to MultiPlus via `VE.Bus` RJ45.
- Short AC-in shore-charge test passed at household-source current limits: about `1294W` shore input and about `54.3V x 21.6A` battery charging in bulk.
- MultiPlus LiFePO4 charge profile is programmed/owner-verified by supervised first-battery behavior. Owner reports the Orion/ `12V` buffer path now charges and operates correctly. Hold open: permanent-path shore dead checks/first-use acceptance, full-bank charge/rest/match/parallel closeout, AC-out branch/GFCI, alternator commissioning, Cerbo hard-mounting, and final strain-relief/abrasion-control.

Current Physical Installation Snapshot (`2026-08-03`)

- The electrical module is hard-mounted through the finished Lonseal/plywood floor into registered truck-bed hardpoints and tied into the hard-mounted Bench/Galley extrusion structure. Owner reports the integrated assembly is extremely stiff; remaining mobile restraint is battery/cooler capture plus terminal/cable protection.
- The driver-rear shore inlet and exterior-to-module cable route are installed. One accessible, enclosed, conductor/gauge-rated three-wire `L/N/PE` splice into the AC-input side remains before dead checks and controlled energization.
- All three batteries' positive and negative `2/0 AWG` branch cables are cut, lugged, heat-shrunk, and landed at the battery-side busbars. Battery 1 completed the corrected isolated charge cycle, Battery 2 resumed and reached absorption on `2026-08-03` , and Battery 3 remains pending; keep all three isolated until their rested voltages are matched within `0.1V` before paralleling.
- The `12V` buffer-battery/Orion branch is owner-reported operational. Keep its negative direct to the fuse-panel main negative stud, not through `SW-12V-BATT` ; the positive path remains `F-11 -> SW-12V-BATT -> panel main +` .
- Owner report `2026-08-03` : the ICECO appliance lead is landed on `12V-02` , protected by a `15A` blade fuse, and switched. Its appliance pigtail is `16 AWG` ; exposed positive switch terminals still require individual insulation and temporary hard mounting before normal

powered work. Verify connector polarity and voltage drop before appliance acceptance.

- Owner report `2026-08-05`: the pump, ICECO/fridge feed, and KUS sender circuit are wired; three separate `12 AWG / 15A` USB-PD branches are routed (`1x` Desk and `2x` Galley); and the Desk and Galley GFCIs are wired on separate breakers. Commissioning remains open for exact fuse/slot labels, polarity/load tests, KUS configuration/sanity check, AC LINE/LOAD/PE proof, and GFCI trip/reset. The third PD branch's actual source slot must be audited and labeled rather than inferred from this document.

Orion fuse discriminator — one input fuse only

- Lynx Slot 4 / `F-05`: **Orion `48V` input positive**, one verified `40A` MEGA (`58VDC` minimum under the locked `56.8V` charge ceiling; Victron `CIP138040020 40A/80V` is the replacement fallback) feeding the existing `6 AWG` directly. This is the final single input fuse.
- `F-06`: **retired standalone Orion input-fuse position**. Do not install MIDI, FKS/ATO, DIN, or a second fuse after Slot 4.
- Lynx Slot 2 / `F-03`: **MPPT branch**, `60A/80V` MEGA. If the physically X-marked/misrated Lynx fuse is in Slot 2, replace it with the purchased Victron `60A/80V` MEGA; it is not the Orion input fuse.
- `F-07`: **Orion `12V` output positive**, `60A/80V` MEGA in the separate Victron holder near the Orion.
- `F-11`: **`12V` buffer-battery positive**, `100A` ANL near the battery, upstream of `SW-12V-BATT`.
- No `32V`-rated fuse is acceptable on any energized `48V` branch. Identify the physical X-marked fuse by Lynx slot before replacing or discarding it.

Length Estimation Defaults Used In This Pass

1. Cabinet internal interconnect default: `2.5 ft` one-way (`ASSUMED`).
2. Cabinet-to-near load branch default: `8 ft` one-way (`ASSUMED`).
3. Cabinet-to-far load branch default: `12 ft` one-way (`ASSUMED`).
4. AC branch to receptacle chain default: `15 ft` one-way per branch leg (`ASSUMED`).
5. Policy lock: use the smallest gauge that meets current and voltage-drop targets; do not auto-upsize, but flag warnings when margin is tight.
6. Parallel battery bank lock: keep each battery's **total positive + negative path resistance** similar. Equal positive-only length is not required if the positive-length differences are offset by negative-length differences; do not add unnecessary 2/0 cable coils solely for cosmetic equality.

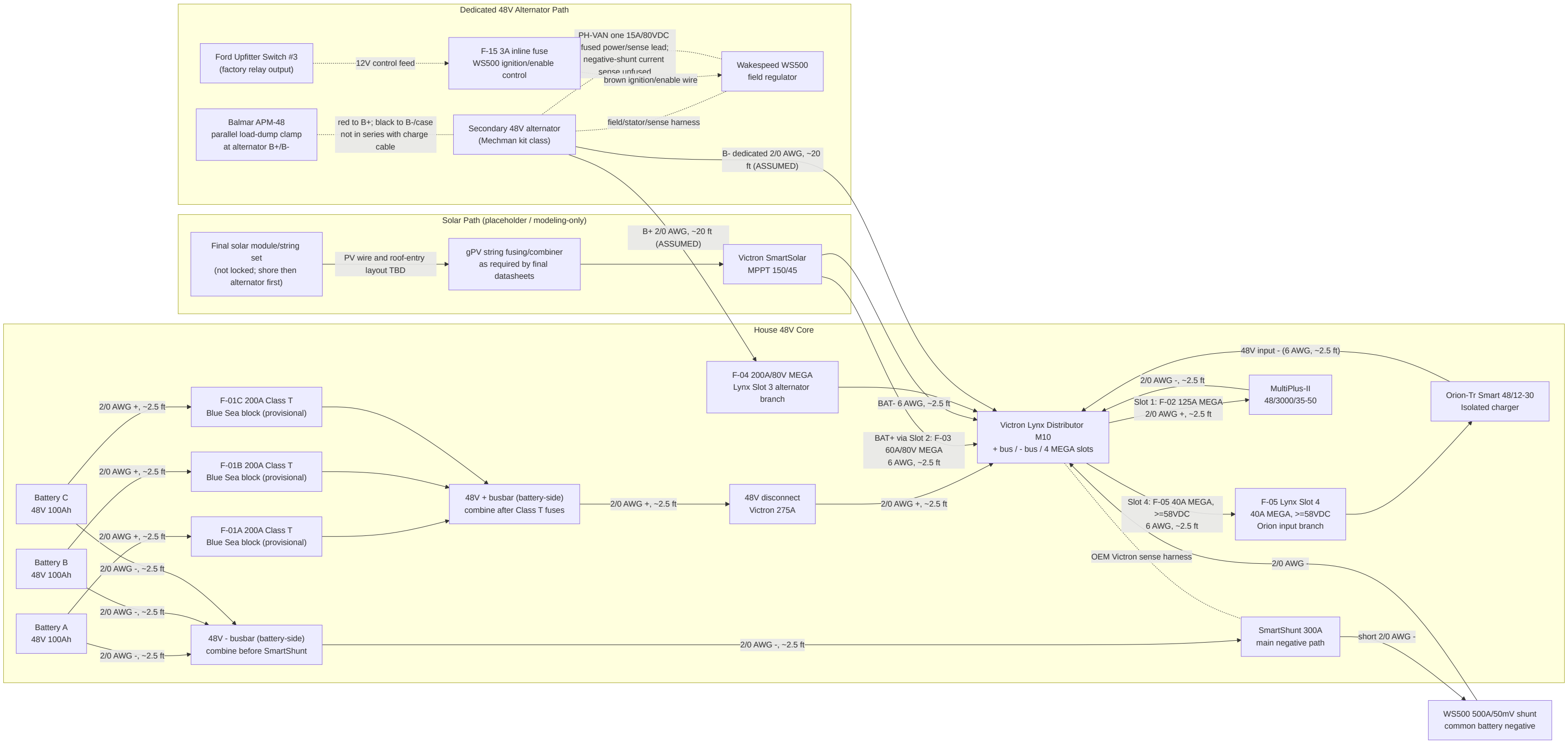
Battery Fuse/Wire Recalculation Basis (2026-02-18 + 2026-03-19 sync)

- Scope in this pass is limited to battery-side and major `48V` trunk paths (`C-01` through `C-15`).
- Provisional battery listing inputs used: `51.2V 100Ah`, `<=200A` current limit per battery.
- Conservative sizing factors used in this pass:

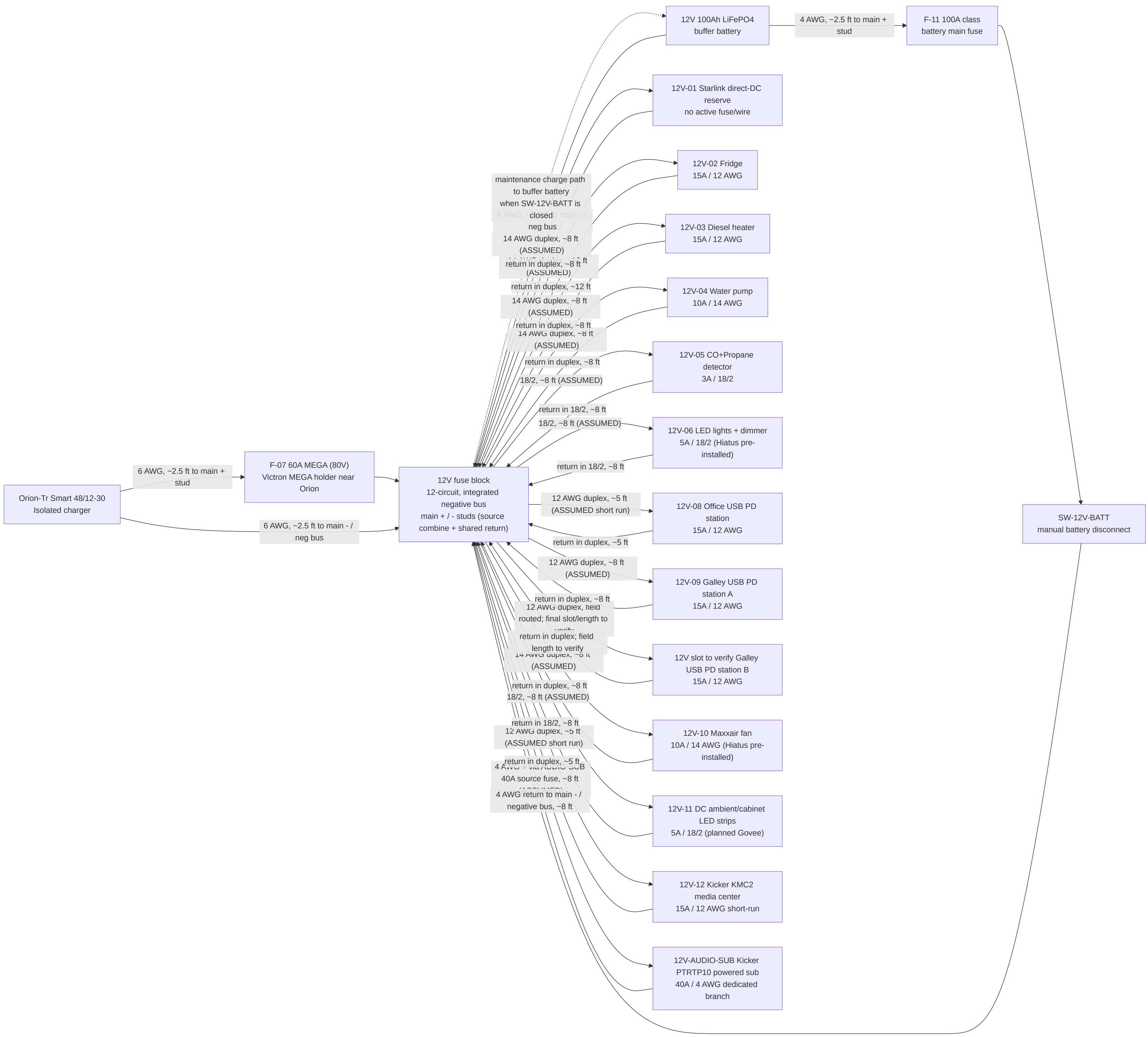
1. Parallel-sharing factor `K_share = 1.5`
2. Continuous margin factor `K_cont = 1.25`

- Current envelope used for battery-discharge branch sizing: `I_total = F-02 + F-05 = 125A + 40A = 165A`.
- Per-battery design current: `I_batt_design = (165A / 3) * 1.5 = 82.5A`.
- Continuous-adjusted minimum battery branch fuse threshold: `82.5A * 1.25 = 103.1A`.
- Provisional battery branch fuse selection: `F-01A/B/C = 200A Class T`, constrained by the provisional battery `<=200A` current-limit listing.
- Final lock gate: validate true `51.2V` battery datasheet/manual current and terminal limits before permanent fuse lock; if lower limits are confirmed, move to `175A`.
- Cable procurement remains estimate-based until CAD/field run lengths are frozen. This pass sets a no-padding `2/0` estimate baseline of `77.5 ft` total (`42.5 ft` red, `35.0 ft` black), including the dedicated alternator migration path.

Complete Power Topology (48V Core + Charge Sources)



12V Distribution Topology (Shared Junction With Buffer Battery)

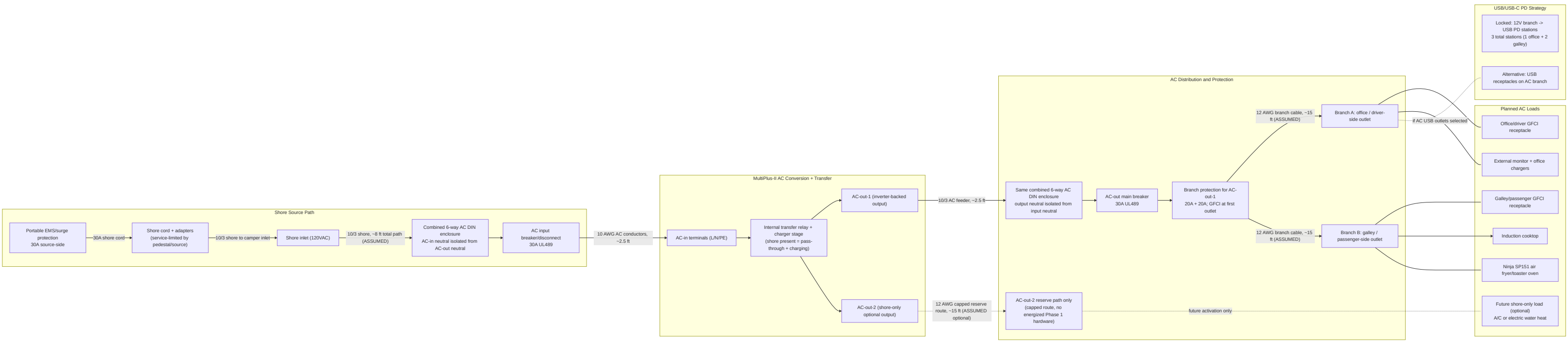


12V Operating Intent (Locked)

- Orion-Tr Smart `48/12-30` is the primary `12V` charger/feed source.
- `SW-12V-BATT` is **normally closed** in operation; open is service/isolation mode only.
- With `SW-12V-BATT` closed, Orion output at the fuse-block main `+` stud maintains/charges the `12V` buffer battery through the shared-junction path.
- With `SW-12V-BATT` open, the buffer battery is isolated from the main `+` stud, but the fuse-panel bus can remain powered from the Orion while the Orion is enabled. For service isolation, turn active loads off, disable the Orion and confirm off/not charging, then open `SW-12V-BATT` and meter the bus before work.
- Replace the Orion's always-on L-H jumper with a maintained SPST dry-contact remote switch. Startup closes `SW-12V-BATT` before enabling the Orion; normal shutdown disables the Orion before opening the battery switch.
- The buffer battery remains in the active operating path during normal use and is intended to absorb transients/peaks on the `12V` rail.
- The fuse block is the `12V` junction device in this baseline: main `+` stud is the source-combine point, and the integrated negative bus/main `-` is the shared return point.
- Do not solder-splice high-current source conductors; terminate with crimped lugs on rated studs/junction hardware.

- Hiatus pre-installed 12V branches are tracked here as `12V-06` (factory LED lights with dimmer) and `12V-10` (Maxxair fan); verify final installed branch labeling during electrical audit.
- Planned Govee/ambient strip lighting is a separate branch (`12V-11`) and not part of the Hiatus factory `12V-06` lighting circuit.
- DC-first lighting intent: keep ambient/cabinet strip lighting on `12V-11` so nighttime lighting does not require inverter operation.
- Camper audio intent: Kicker `46KMC2` source unit uses a `15A` `12V-12` branch from the fuse block, while Kicker `49PTRTP10` powered sub uses a separate `40A` fused `4 AWG` branch from the 12V source/junction rather than a normal blade-fuse panel branch. Bass peaks may exceed the Orion `30A` continuous feed and are supported by the `12V` buffer battery for short durations.

AC Path Topology (Shore + Inverter Output, Full Hierarchy)



AC Operating Behavior (Expected)

- Shore present: MultiPlus transfer relay closes, AC-in is passed to AC-out paths, and charger stage charges the `48V` bank.
- Shore absent: MultiPlus transfers to inverter mode and powers `AC-out-1` from battery; `AC-out-2` drops by design.
- AC-in hardware is `30A` (`source/adapters -> portable EMS -> shore cord -> L5-30 inlet -> 30A breaker -> 10 AWG AC-in conductors`); set MultiPlus input current limit to actual source (`15A` , `20A` , or `30A`) to avoid pedestal/source breaker trips.
- Initial battery charging may use AC-in-only mode with AC-out branch loads disconnected.

AC Safety/Protection Chain (What Must Exist)

- Upstream shore protection chain before MultiPlus AC-in: shore source/adapters -> portable EMS -> shore cord -> shore inlet -> `30A` AC input breaker/disconnect.
- AC-out protection chain: MultiPlus AC-out-1 -> `10/3` feeder -> `30A` AC-out main breaker -> `20A` branch breakers -> GFCI receptacles.
- Single-enclosure DIN architecture with isolated AC-in and AC-out neutral paths and no neutral mixing.
- Continuous equipment grounding path from shore inlet through MultiPlus and branch circuits. The MultiPlus external `M6 PE` lug gets at least `4 mm²` , selected as `10 AWG` green stranded copper, to a verified truck-chassis bond point; the aluminum shell gets a separate corrosion-compatible bond into the same equipment-ground network and is not the only PE path.
- Neutral/ground handling follows the MultiPlus internal relay: AC-out neutral bonds to chassis in inverter mode and opens that bond when shore is accepted. Do not add an always-bonded downstream neutral-ground bond in branch receptacle wiring.
- Keep PE and DC return roles separate: no MultiPlus case/PE jumper to Lynx or `12V` negative. Confirm whether the Mechman is isolated- or case-grounded and prove no chassis path bypasses the SmartShunt before alternator commissioning.

AC Reference Basis (Manufacturer Guidance)

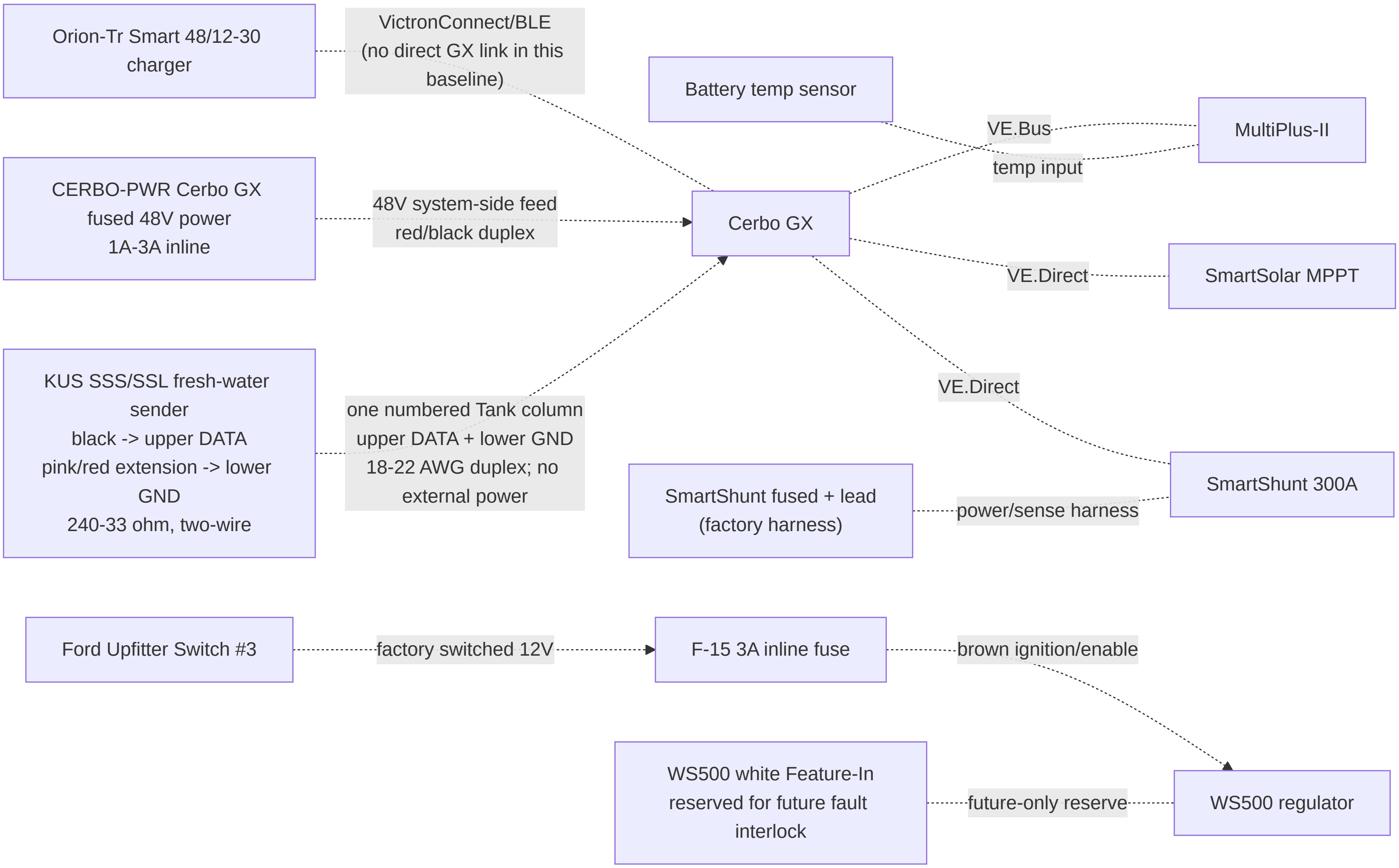
- Victron MultiPlus-II `120V` installation guidance (`AC-in` breaker sizing, UL943-class residual-current protection on outputs, and AC-out-2 shore-only behavior): `https://www.victronenergy.com/media/pg/MultiPlus-II_120V/en/installation.html`
- Victron MultiPlus-II datasheet (`48/3000/35-50` baseline model reference): `https://www.victronenergy.com/upload/documents/Datasheet-MultiPlus-II-inverter-charger-120V-EN.pdf`

AC/USB Baseline Locked For BOM

- Shore interface: `30A` RV-style inlet baseline with adapter kit for `15A` / `20A` hookups.
- AC input protection: portable EMS + combined DIN enclosure + `30A` AC input breaker/disconnect upstream of MultiPlus AC-in.
- AC-out-1 distribution: `30A` AC-out main plus two protected `20A` branches with GFCI-at-first-outlet strategy.
- Receptacle plan: `2` first-in-chain `120V` GFCI receptacles remain required, one per active branch: Branch A = office/driver side, Branch B = galley/passenger side. Owner reopened a small number of downstream receptacles on `2026-08-05` ; final count/locations remain field-fit gated and must continue from each first GFCI's `LOAD` terminals with `12 AWG` conductors, accessible listed boxes, and full-chain trip testing.
- USB charging plan: `3` DC-fed USB PD station assemblies on separate `12V` branches (`1` office, `2` Galley), each standardized to `15A / 12 AWG` . The third branch's actual fuse-panel slot/label is a field-audit item.

- AC-out-2 remains reserve-only in Phase 1 (labeled capped route; no energized branch hardware procured).

Monitoring and Control Topology



Fuse and Switch Housing Map (Where Each Item Is Physically Housed)

Item ID	Item value/type	Housing method	Location
`F-01A`	`200A Class T` (provisional)	Blue Sea Class T fuse block (`110A-200A` family)	Battery compartment near Battery A `+`
`F-01B`	`200A Class T` (provisional)	Blue Sea Class T fuse block (`110A-200A` family)	Battery compartment near Battery B `+`
`F-01C`	`200A Class T` (provisional)	Blue Sea Class T fuse block (`110A-200A` family)	Battery compartment near Battery C `+`
`F-02`	`125A MEGA`	Lynx integrated slot holder	Lynx Slot 1
`F-03`	`60A MEGA` (`80V` Victron replacement stock)	Lynx integrated slot holder	Lynx Slot 2
`F-04`	`200A/80V MEGA`	Lynx integrated slot holder	Lynx Slot 3 (dedicated alternator branch)
`F-05`	`40A` MEGA, body-marked `>=58VDC`; Victron `CIP138040020 40A/80V` replacement fallback	Lynx integrated slot holder	Lynx Slot 4; Orion `48V` input feeder
`F-06`	Not installed / retired standalone input-fuse position	No holder	MIDI/FKS/DIN concepts superseded; do not stack after `F-05`

F-07	60A MEGA (80V Victron replacement stock)	Victron MEGA fuse holder	Electrical cabinet at Orion 12V + source end
F-09A/B/C	15A gPV each	10x38 touch-safe fuse holders in PV combiner	Roof-entry combiner enclosure
F-10	Per branch (ATO/ATC)	Integrated blade sockets in generic 12V fuse block	Electrical cabinet
AUDIO-HU / 12V-12	15A source/head-unit branch; KMC2 harness also contains a 15A ATM fuse	12V fuse block branch plus KMC2 harness fuse	Electrical cabinet to driver-side DC shelf/source face
AUDIO-SUB	40A external fuse for Kicker PTRTP10 powered sub branch	Inline/MRBF/AFS/ANL-class holder matched to selected 4 AWG kit; use 40A, not a generic 100A kit fuse	Within about 18 in of the 12V source takeoff feeding the powered sub
F-11	100A class (12V buffer battery main)	Sealed inline MIDI/AMI/ANL holder	Within ~`7"` of 12V buffer battery positive post
F-12/F-13-PHVAN	Mersen ATDR15, 15A/300VDC time-delay Class-CC fuse	Mersen USCC1 600VDC touch-safe holder in a separate small DC DIN enclosure	Immediately beside the Lynx, fed from the F-04 alternator/load-side stud through short 14 AWG pigtails; sealed splice to 16 AWG PH-VAN red
CERBO-PWR	1A-3A Cerbo GX power fuse	Small inline holder rated for the 48V bank maximum	Electrical cabinet near 48V system-positive takeoff; system side of disconnect preferred for bench shutdown
WS500 current-sense pair	No fuse; common battery-negative shunt with purple/high on battery/SmartShunt side and grey/low on Lynx/system side	Twist pair if extended; route away from noise; configure Shunt at Battery	Wakespeed shunt in series after the SmartShunt
F-15	3A WS500 ignition/enable control fuse	Sealed inline ATC/ATO holder; 12V control circuit	Near Ford upfitter blunt-cut wire / WS500 control-wire handoff
SW-12V-BATT	Manual battery disconnect switch	Sealed rotary DC switch body	Electrical cabinet near 12V fuse-block main + stud for service access
OEM-SHUNT	External Victron-supplied inline fuse in red SmartShunt Vbatt+ cable	Inline holder in supplied red cable	Prefer battery-side positive if SOC continuity is desired with the main disconnect open; system side is acceptable for zero disconnect-off parasitic draw

Retired from active architecture:

- Obsolete pre-Mechman charger/fuse paths are removed from active board layout and labels.

Conductor Schedule (Start-to-Finish)

Segment ID	Circuit segment	Nominal voltage	Current basis	Overcurrent protection	Planned wire gauge	Estimated one-way length (this pass)
C-01	Battery A + -> F-01A	48V	Battery branch, fuse-limited	F-01A 200A provisional	2/0 AWG	2.5 ft planning placeholder; balance total loop path
C-02	Battery B + -> F-01B	48V	Battery branch, fuse-limited	F-01B 200A provisional	2/0 AWG	2.5 ft planning placeholder;

						balance total loop path
C-02C	Battery C + -> F-01C	48V	Battery branch, fuse-limited	F-01C 200A provisional	2/0 AWG	2.5 ft planning placeholder; balance total loop path
C-03	Class T outputs -> battery-side 48V + busbar -> disconnect input	48V	Combined trunk current	F-01A/B/C	2/0 AWG each branch	2.5 ft each branch (ASSUMED, 4 conductors in rollup)
C-04	Disconnect output -> Lynx + bus	48V	Aggregate discharge design current (165A = F-02 125A + Orion F-05 40A)	Upstream Class T fuses	2/0 AWG	2.5 ft (ASSUMED)
C-05	Battery negatives -> battery-side 48V - busbar -> SmartShunt battery side	48V	Mixed-path rollup: 3x battery-negative branches at 82.5A design each + NEGBUS_TO_SHUNT trunk at 165A aggregate	N/A (main negative path)	2/0 AWG each branch	2.5 ft each branch (ASSUMED, 4 conductors in rollup)
C-06	SmartShunt load side -> Lynx - bus	48V	Aggregate return current	N/A	2/0 AWG	2.5 ft (ASSUMED)
C-06A	Lynx positive tap -> SmartShunt positive sense/power lead	48V	Shunt electronics supply (very low current)	Factory inline fuse in OEM harness	OEM harness lead	2.5 ft (ASSUMED)
C-07	Lynx Slot 1 (F-02) -> MultiPlus DC+	48V	Inverter branch, fuse-limited	F-02 125A	2/0 AWG (manual minimum AWG 1 on short runs)	2.5 ft (ASSUMED)
C-08	MultiPlus DC- -> Lynx - bus	48V	Inverter return current	F-02 protects paired positive	2/0 AWG	2.5 ft (ASSUMED)
C-09	MPPT BAT+ -> Lynx Slot 2 (F-03)	48V	Controller output (45A max)	F-03 60A/80V Victron row 188	6 AWG	2.5 ft (ASSUMED)
C-10	MPPT BAT- -> Lynx - bus	48V	Controller return current	F-03 protects paired positive	6 AWG	2.5 ft (ASSUMED)
C-11	Secondary alternator B+ -> APM-48 -> Lynx Slot 3 (F-04)	48V	Published alternator output reaches about 145.7A	F-04 200A/80V	2/0 AWG	20 ft (ASSUMED, one-way)
C-12	Secondary alternator B- -> Lynx - bus (dedicated return)	48V	Alternator branch return current	F-04 paired	2/0 AWG	20 ft (ASSUMED, one-way)
C-13/C-14	Lynx Slot 4 (F-05) -> Orion 48V + input	48V	Orion feeder, fuse-limited	F-05 40A MEGA, >=58VDC	Existing 6 AWG direct; 10 AWG adequate if ever replaced	2.5 ft (ASSUMED)
C-15	Orion 48V - input -> Lynx - bus	48V	Orion input return current	Orion input positive protection paired	6 AWG	2.5 ft (ASSUMED)

C-18	Orion 12V + -> F-07 -> 12V fuse block main + stud	12V	Charger output path (30A continuous, 60A fuse)	F-07 60A	6 AWG planned (8 AWG minimum per Orion table)	2.5 ft (ASSUMED)
C-19	Orion 12V - -> 12V fuse block integrated - bus / main - stud	12V	Charger output return	F-07 protects paired positive	6 AWG	2.5 ft (ASSUMED)
C-19A	12V buffer battery + -> F-11 -> SW-12V-BATT -> 12V fuse block main + stud	12V	Buffer source path and service isolation path	F-11 100A class	4 AWG planned	2.5 ft (ASSUMED)
C-19B	12V buffer battery - -> 12V fuse block integrated - bus / main - stud	12V	Buffer battery return path	N/A (paired with C-19A)	4 AWG planned	2.5 ft (ASSUMED)
C-20	Reserved 12V panel -> future Starlink direct-DC conversion	12V	Not active; Standard 4 X remains on its supplied AC power path	No fuse installed until exact converter/input is selected	No conductor committed	Pull string/conduit only if useful while access is open
C-21	12V panel -> Fridge	12V	Branch load	F-10 15A	12 AWG duplex	12 ft (ASSUMED, far-load branch; upsize avoids the prior voltage-drop warning)
C-22	12V panel -> LF Bros diesel-heater harness	12V	Startup/cooldown branch; keep energized through controller-commanded shutdown	F-10 15A (retain any supplied harness fuse; verify coordination)	12 AWG duplex	8 ft (ASSUMED; measure before final cable landing)
C-23	12V panel -> Water pump	12V	Branch load	F-10 10A	14 AWG duplex	8 ft (ASSUMED, near-load branch)
C-24	12V panel -> CO + propane detector	12V	Branch load	F-10 3A	18/2	8 ft (ASSUMED, near-load branch)
C-25	12V panel -> LED lights + dimmer (Hiatus pre-installed)	12V	Branch load	F-10 5A	18/2	8 ft (ASSUMED, near-load branch)
C-26	48V system positive/negative -> CERBO-PWR -> Cerbo GX power input	48V	Cerbo electronics feed (~3W)	CERBO-PWR 1A-3A inline fuse	18 AWG red/black duplex acceptable	2.5 ft (ASSUMED, cabinet internal)
C-27	PV strings -> F-09 combiner -> MPPT PV input	PV string voltage (3S)	String current + combiner output current	F-09A/B/C 15A each	10 AWG PV wire	12 ft trunk + 3x8 ft string legs (ASSUMED)
C-28	Shore source/adapters -> portable EMS ->	120VAC	Source-limited shore current (adapter-constrained at source)	30A AC input breaker/disconnect baseline with	10/3 shore feed to	8 ft (ASSUMED)

	shore cord -> shore inlet -> combined AC DIN enclosure / AC input breaker			source-current-limit settings policy	inlet/AC-in area	
C-29	AC input breaker/disconnect -> MultiPlus AC-in	120VAC	MultiPlus AC input current (30A hardware basis)	Upstream 30A AC breaker/disconnect (C-28)	10 AWG stranded AC conductors	2.5 ft (ASSUMED, cabinet internal)
C-30	MultiPlus AC-out-1 -> combined AC DIN enclosure / AC-out main breaker	120VAC	Inverter-backed AC-out feeder current (30A system cap)	30A AC-out main breaker	10/3 stranded AC feeder	2.5 ft (ASSUMED, cabinet internal)
C-31	Branch A -> office/driver GFCI receptacle	120VAC	Branch load (office monitor/chargers/general outlet use)	20A branch breaker + GFCI receptacle	12/3 stranded AC branch cable	15 ft (ASSUMED, branch leg default)
C-32	Branch B -> galley/passenger GFCI receptacle	120VAC	Branch load (galley/general high-draw outlet; induction/Ninja SP151 sequenced)	20A branch breaker + GFCI receptacle	12/3 stranded AC branch cable	15 ft (ASSUMED, branch leg default)
C-33	MultiPlus AC-out-2 (reserve-only) -> capped route for future shore-only branch	120VAC	N/A in Phase 1 (route reserved only)	N/A in Phase 1 (no energized branch hardware)	12 AWG stranded AC conductors (reserve path only)	15 ft (ASSUMED, reserve route default)
C-34	12V panel -> USB PD station branch (office zone)	12V	One Acegoo 118W station	F-10 branch fuse (15A)	12 AWG duplex baseline	5 ft (ASSUMED, short-run requirement)
C-35	12V panel -> USB PD station branch (galley zone)	12V	Galley charging branch (65W class USB-C plus USB-A/C loads)	F-10 branch fuse (15A)	12 AWG duplex baseline	8 ft (ASSUMED; standardized with the office PD/fridge rough-in for voltage-drop margin)
C-36	12V panel -> Maxxair fan (Hiatus pre-installed)	12V	Roof ventilation branch	F-10 branch fuse (10A)	14 AWG duplex baseline	8 ft (ASSUMED, near-load branch)
C-37	12V panel -> DC ambient/cabinet LED strips (planned Govee)	12V	Branch load	F-10 branch fuse (5A)	18/2 baseline	8 ft (ASSUMED, near-load branch)
C-38	F-04 alternator/load-side stud -> Mersen USCC1/ATDR15 -> WS500 PH-VAN red lead	48V bank	Combined regulator-power / positive-voltage-sense feed	F-12/F-13-PHVAN (15A)	Short 14 AWG pigtails, sealed transition to 16 AWG harness	Local beside Lynx/F-04
C-39	Retired separate WS500 positive-sense conductor	N/A	Not installed with confirmed PH-VAN harness	None	N/A	N/A

C-40	WS500 current-sense pair to common battery-negative shunt: purple/high battery/SmartShunt side, grey/low Lynx/system side	low-current sense	Net battery-current feedback	No fuse; twist pair if extended; configure Shunt at Battery	Harness lead	Local near house board
C-41	Ford Upfitter #3 output -> F-15 -> WS500 brown ignition/enable wire	12V control lead	Manual alternator-enable signal only	F-15 (3A)	16 AWG TXL/GXL	6 ft (ASSUMED)
C-42	12V panel -> Kicker 46KMC2 media center/source unit	12V	Source/head-unit branch (15A max; KMC2 manual shows 15A ATM)	F-10/AUDIO-HU 15A branch plus KMC2 harness fuse	12 AWG duplex if kept near 5 ft; use 10 AWG if longer	5 ft (ASSUMED, driver-side DC shelf)
C-43	12V source/main + stud -> AUDIO-SUB source fuse -> Kicker 49PTRTP10 powered sub +	12V	Powered sub branch; PTRTP10 manual external fuse value	AUDIO-SUB 40A	4 AWG tinned/OFC	8 ft (ASSUMED; shorten by placing sub near 12V junction)
C-44	Kicker 49PTRTP10 powered sub - -> 12V negative bus/main - stud	12V	Powered sub return	AUDIO-SUB protects paired positive	4 AWG tinned/OFC	8 ft (ASSUMED, paired with positive)
C-45	KMC2 remote turn-on output -> PTRTP10 remote input	12V low-current control	Remote amp/sub enable	KMC2/source branch protected	18 AWG	8 ft (ASSUMED)
C-46	KMC2 RCA line-out -> PTRTP10 RCA input	low-level audio signal	Subwoofer signal	N/A	shielded 2-channel RCA	measured route (~4m planning cable)
C-47L/R	KMC2 front speaker outputs -> left/right Kicker CSC67 speakers	speaker-level audio	One 4 ohm speaker per KMC2 channel	KMC2 internal protection / AUDIO-HU source branch	16 AWG marine speaker wire	10-12 ft each (ASSUMED)
C-48	KUS SSS/SSL sender black signal + pink return -> matching signal/negative pair in Cerbo GX MK2 Tank 1 column	Cerbo low-current resistive-sender excitation	Fresh-water level, US 240-30 ohm	No external fuse or power feed; do not share chassis return	18-22 AWG tinned duplex, ferrules at Cerbo, sealed pigtail splices at sender	Measure after tank/Cerbo endpoints are installed (ASSUMED <10 ft)
C-49	12V panel -> second Galley USB PD station	12V	Third independent PD branch	F-10 branch fuse (15A); exact panel slot/label to verify in field	12 AWG duplex	Owner reports routed 2026-08-05; measure/record final length

Wiring Validation Worksheet (Estimate Pass, 2026-02-18)

Calculation basis for drop screening:

1. `V_drop = I * (2 * L_one_way * R_per_ft)`
2. Resistance basis used in this pass (`ohm/ft`): `2/0=0.0000779` , `6 AWG=0.0003951` , `4 AWG=0.0002485` , `12 AWG=0.001588` , `14 AWG=0.002525` , `18/2=0.006385` , `10 AWG=0.000999` .
3. Design targets: `<=2%` on major `48V` trunks, `<=3%` on planned `12V` /AC branches.
4. `C-05` is a rollup row that includes both branch and trunk return paths; voltage-drop screen shown is the conservative interim worst-case (`155A`) within that rollup.

Circuit ID	From	To	Fuse	Current basis	Gauge	Estimated one-way length	Voltage drop %	B
C-01	Battery A +	F-01A	F-01A 200A	77.5A interim design branch share	2/0 AWG	2.5 ft planning placeholder; field layout may use unequal positive lengths if total loop path is balanced	0.059% @ 51.2V planning placeholder	Ro re
C-02	Battery B +	F-01B	F-01B 200A	77.5A interim design branch share	2/0 AWG	2.5 ft planning placeholder; field layout may use unequal positive lengths if total loop path is balanced	0.059% @ 51.2V planning placeholder	Ro re
C-02C	Battery C +	F-01C	F-01C 200A	77.5A interim design branch share	2/0 AWG	2.5 ft planning placeholder; field layout may use unequal positive lengths if total loop path is balanced	0.059% @ 51.2V planning placeholder	Ro re
C-03	Class T load studs	48V + bus / disconnect input	F-01A/B/C	77.5A interim per branch	2/0 AWG	2.5 ft each (x4 conductors)	0.059% @ 51.2V	Ro re
C-04	Disconnect output	Lynx + bus	Upstream Class T	155A interim / 145A final aggregate	2/0 AWG	2.5 ft	0.12% @ 51.2V interim	Ro re
C-05	Battery - branches	SmartShunt battery side via 48V - bus	N/A	77.5A per battery-negative branch; row rollup also includes one 155A interim	2/0 AWG	2.5 ft each (x4 conductors)	0.12% @ 51.2V (worst-case rollup)	Ro bl

				trunk (NEGBUS_TO_SHUNT)				
C-06	SmartShunt load side	Lynx - bus	N/A	155A interim / 145A final aggregate return	2/0 AWG	2.5 ft	0.12% @ 51.2V interim	Ro bl
C-06A	Lynx positive tap	SmartShunt sense/power lead	OEM inline fuse	OEM harness current	OEM harness	2.5 ft	N/A (low-current OEM lead)	Ro ha
C-07	Lynx Slot 1 DC+	MultiPlus DC+	F-02 125A	125A	2/0 AWG	2.5 ft	0.10% @ 51.2V	Ro re
C-08	MultiPlus DC-	Lynx - bus	F-02 paired	125A	2/0 AWG	2.5 ft	0.10% @ 51.2V	Ro bl
C-09	MPPT BAT+	Lynx Slot 2	F-03 60A (80V Victron row 188)	45A	6 AWG	2.5 ft	0.17% @ 51.2V	Ro re
C-10	MPPT BAT-	Lynx - bus	F-03 paired	45A	6 AWG	2.5 ft	0.17% @ 51.2V	Ro bl
C-11	Secondary alternator B+	Lynx Slot 3 via APM-48	F-04 200A/80V	145.7A published-curve screen	2/0 AWG	20 ft	<0.80% @ 58.4V	Ro re
C-12	Secondary alternator B-	Lynx - bus (dedicated return)	F-04 paired	145.7A published-curve screen	2/0 AWG	20 ft	<0.80% @ 58.4V	Ro bl
C-13/C-14	Lynx Slot 4	Orion 48V +	F-05 40A MEGA, >=58VDC	40A fuse basis	Existing 6 AWG direct	2.5 ft	0.16% @ 51.2V	Ro re
C-15	Orion 48V -	Lynx - bus	Orion input positive protection paired	40A fuse basis	6 AWG	2.5 ft	0.16% @ 51.2V	Ro bl
C-18	Orion 12V +	Fuse block main + stud	F-07 60A/80V Victron row 188	30A	6 AWG	2.5 ft	0.49% @ 12V	Ro re
C-19	Orion 12V -	Fuse block integrated - bus / main - stud	F-07 paired	30A	6 AWG	2.5 ft	0.49% @ 12V	Ro bl
C-19A	Buffer battery +	Fuse block main + stud (via F-11/SW)	F-11 100A	50A design	4 AWG	2.5 ft	0.52% @ 12V	Ro re
C-19B	Buffer battery -	Fuse block integrated - bus / main - stud	N/A	50A design	4 AWG	2.5 ft	0.52% @ 12V	Ro bl
C-20	12V fuse panel	Future Starlink direct-DC conversion	None active	N/A	No conductor committed	N/A	N/A	Co BC

C-21	12V fuse panel	Fridge	F-10 15A	7A	12 AWG duplex	12 ft	2.23% @ 12V	Ro fie du
C-22	12V fuse panel	LF Bros diesel-heater harness	F-10 15A + supplied harness fuse if present	10A startup screen	12 AWG duplex	8 ft assumed	2.12% @ 12V	Fie du
C-23	12V fuse panel	Water pump	F-10 10A	7A	14 AWG duplex	8 ft	2.36% @ 12V	Ro AW
C-24	12V fuse panel	CO+propane detector	F-10 3A	0.2A	18/2	8 ft	0.17% @ 12V	Ro
C-25	12V fuse panel	LED lights + dimmer (Hiatus pre-installed)	F-10 5A	5A	18/2	8 ft	4.26% @ 12V	Ro
C-26	48V system feed	Cerbo GX	CERBO-PWR 1A-3A	~3W	18 AWG duplex acceptable	2.5 ft	N/A (low-current electronics feed)	Lo in ro
C-27	PV strings/combiner	MPPT PV input	F-09A/B/C 15A	30A trunk screen	10 AWG PV	12 ft trunk + 3x8 ft string legs	0.72% @ 100V trunk screen	Ro AV ec
C-28	Shore inlet path	AC input breaker/disconnect	Source-limited AC OCP	30A hardware basis	10/3	8 ft	0.40% @ 120VAC	Ro sh in
C-29	AC input breaker/disconnect	MultiPlus AC-in	Upstream AC OCP	30A hardware basis	10 AWG AC	2.5 ft	0.12% @ 120VAC	Ro sh in
C-30	MultiPlus AC-out-1	AC-out main breaker	30A AC-out main	30A hardware basis	10/3	2.5 ft	0.12% @ 120VAC	Ro sh in
C-31	Branch A	Office/driver receptacle chain	20A branch OCP	20A	12 AWG AC	15 ft	0.79% @ 120VAC	Ro AW
C-32	Branch B	Galley/passenger receptacle chain	20A branch OCP	20A	12 AWG AC	15 ft	0.79% @ 120VAC	Ro AW
C-33	MultiPlus AC-out-2	Reserve-only capped route	N/A in Phase 1	N/A	12 AWG AC (reserve path only)	15 ft	0.60% @ 120VAC (future-use screen)	N/ pr Ph
C-34	12V fuse panel	Office USB PD station	F-10 15A	15A design cap	12 AWG duplex	5 ft	1.99% @ 12V	Ro AW br
C-35	12V fuse panel	Galley USB PD station	F-10 15A	8A expected	12 AWG duplex	8 ft	1.69% @ 12V	Ro AW br
C-49	12V fuse panel	Second Galley USB PD station	F-10 15A; slot to verify	15A design cap	12 AWG duplex	Field routed; measure	Recalculate from measured length	Fie

C-36	12V fuse panel	Maxxair fan (Hiatus pre-installed)	F-10 10A	4A expected	14 AWG duplex	8 ft	1.35% @ 12V	Ro AW
C-37	12V fuse panel	DC ambient/cabinet LED strips (planned Govee)	F-10 5A	5A design cap	18/2	8 ft	4.26% @ 12V	Ro
C-38	F-04 alternator/load-side stud	WS500 PH-VAN short red lead	F-12/F-13-PHVAN Mersen ATDR15 15A/300VDC	combined regulator-power / voltage-sense feed at 48V bank voltage	14 AWG pigtails to 16 AWG harness	Local	N/A (harness-limited)	Ac 17 US
C-39	Retired separate positive-sense source	Not installed	None	superseded by confirmed PH-VAN combined lead	N/A	N/A	N/A	In 32
C-40	Common battery-negative Wakespeed shunt: purple/high battery/SmartShunt side, grey/low Lynx/system side	WS500 current-sense input	No fuse	low-current sense; twist pair if extended	Harness lead	Local	N/A (harness-limited)	Ha
C-41	Ford Upfitter #3 output	WS500 brown ignition/enable input via F-15	F-15 3A	manual low-current control only	16 AWG TXL/GXL	6 ft	N/A (control circuit)	Ro (u co
C-42	12V panel	Kicker 46KMC2 media center	AUDIO-HU 15A branch + KMC2 15A ATM harness fuse	15A max fuse basis	12 AWG duplex	5 ft	1.99% @ 12V	Ro au
C-43/C-44	12V source/main studs	Kicker 49PTRTP10 powered sub	AUDIO-SUB 40A external source fuse	40A manual fuse basis	4 AWG positive + matching return	8 ft	1.33% @ 12V	Ro au kit
C-45	KMC2 remote output	PTRTP10 remote input	KMC2/source branch protected	low-current remote	18 AWG	8 ft	N/A	Ro au sig
C-46	KMC2 RCA line-out	PTRTP10 RCA input	N/A	low-level signal	shielded RCA	measured route	N/A	Ro au sig
C-47L/R	KMC2 front speaker outputs	Left/right CSC67 speakers	KMC2/source branch protected	one 4 ohm speaker per channel	16 AWG marine speaker wire	10-12 ft each	N/A	Ro au sp

Wire Rollup (No-Padding Purchase Baseline)

Gauge / cable family	Estimated total	Source circuits	BOM row
2/0 AWG red	42.5 ft	C-01, C-02, C-02C, C-03, C-04, C-07, C-11	28

2/0 AWG black	35.0 ft	C-05, C-06, C-08, C-12	28
6 AWG red	7.5 ft	C-09, C-13/C-14, C-18	29
6 AWG black	7.5 ft	C-10, C-15, C-19	29
4 AWG red	10.5 ft (2.5 ft existing + 8 ft audio sub planning branch)	C-19A, C-43	30, 192
4 AWG black	10.5 ft (2.5 ft existing + 8 ft audio sub return)	C-19B, C-44	30, 192
10 AWG pair-equivalent (PV)	placeholder only	C-27 final module/string topology deferred until solar workstream; do not buy combiner/fuse count or cut roof entries from the old 3S3P model	31
14 AWG duplex	16 ft	C-23, C-36	32
18/2	24.0 ft plus separate Cerbo low-current duplex	C-24, C-25, C-37; Cerbo C-26 uses 48V fused low-current duplex	33 / low-current stock
12 AWG AC branch cable	30 ft (C-33 excluded in Phase 1)	C-31, C-32	113
10/3 shore + AC-in/out feed	13 ft	C-28, C-29, C-30	114
12 AWG fridge/heater/USB/audio branch mix	Fridge: 12 ft; heater: 8 ft; USB: 5 ft office + 8 ft Galley A + measured Galley B field run; audio source: 5 ft short-run assumption	C-21, C-22, C-34, C-35, C-49, C-42	116, 193 / field stock
Camper audio signal/speaker/control wiring	RCA measured route, 18 AWG remote, and 16 AWG marine speaker wire runs	C-45, C-46, C-47L/R	193
WS500 harness/sense leads	Included in selected kit/harness set; C-39 is retired under PH-VAN	C-38, C-40	168, 171
16 AWG TXL/GXL control wire	6 ft	C-41	176

Notes:

1. This table is a base estimate only; it intentionally excludes order padding and termination waste.
2. Apply personal order overage at checkout based on actual spool cut increments and routing confidence.
3. Parallel bank balancing is locked by similar total loop resistance per battery. Positive-only leads may differ if the paired negative path offsets the difference; verify with final measured lengths and, if needed, clamp-current checks.

3x Battery Bank Bench-Build Cut List (2/0 AWG)

Purpose: make the bench build orderable without needing final camper run lengths. Treat lengths below as *bench module* lengths only; final install harnesses should be re-cut after layout freeze.

Assumptions:

1. Battery terminals are `M8` (verify your battery stud size before ordering lugs).
2. Class T fuse blocks and battery-side busbars use `3/8"` studs (treat as `M10` lugs unless your specific hardware differs).
3. Lynx Distributor is the `M10` model (main connections `M10`; internal/fuse studs may still be `M8` depending on the position).

Cable ID	Qty	From -> To	Color	Gauge	Est. one-way length	Lug A	Lug B
BATT+_A/B/C	3	Battery + -> Class T block line side	red	2/0	2.5 ft each (ASSUMED)	M8	M10
FUSE_TO_POSBUS_A/B/C	3	Class T block load side -> 48V + busbar	red	2/0	2.5 ft each (ASSUMED)	M10	M10
POSBUS_TO_DISC	1	48V + busbar -> disconnect input	red	2/0	2.5 ft (ASSUMED)	M10	M10

DISC_TO_LYNX+	1	disconnect output -> Lynx + input	red	2/0	2.5 ft (ASSUMED)	M10	M10
BATT-_A/B/C	3	Battery - -> 48V - busbar	black	2/0	2.5 ft each (ASSUMED)	M8	M10
NEGBUS_TO_SHUNT	1	48V - busbar -> SmartShunt battery side	black	2/0	2.5 ft (ASSUMED)	M10	M10
SHUNT_TO_LYNX-	1	SmartShunt load side -> Lynx - input	black	2/0	2.5 ft (ASSUMED)	M10	M10
LYNX_SLOT1_TO_MULTI+	1	Lynx Slot 1 DC+ -> MultiPlus DC+	red	2/0	2.5 ft (ASSUMED)	M8	M8
LYNX_TO_MULTI-	1	Lynx - -> MultiPlus DC-	black	2/0	2.5 ft (ASSUMED)	M8	M8

Locked balancing rule for the `3x` parallel bank:

1. Keep each battery path's total positive + negative loop resistance similar.
2. Equal positive-only leads are not required when the negative lead lengths intentionally offset the positive lead differences.
3. Keep the same cable family, lug geometry, fuse/holder family, and termination quality across all three battery paths.
4. Verify sharing with clamp-current checks under charge/load if final measured path lengths differ materially.

Torque reference (verify against your exact manuals/hardware):

- MultiPlus-II DC terminals: `12 Nm` (`M8` nut) per Victron installation guidance.
- SmartShunt shunt bolts: verify torque for the installed `300A` model per Victron installation guidance.
- Lynx Distributor `M10` model: `M10` nuts `33 Nm` (older serials may be lower), and `M8` nuts `14 Nm` per Victron Lynx installation guidance.

Additional Components Included In Topology Scope

- `48V` disconnect (`275A`)
- Pre-charge resistor (commissioning/soft-charge aid before connecting large DC loads)
- Battery-side `48V +` combine busbar (after Class T fuses)
- Battery-side `48V -` combine busbar (battery-only, before SmartShunt)
- 12V fuse block main `+` stud used as source-combine point (Orion + buffer battery feed)
- 12V fuse block integrated negative bus/main `-` used as shared return point
- 12V buffer battery main fuse (`F-11`) and manual disconnect switch (`SW-12V-BATT`)
- Shore AC inlet + cord/adapter interface hardware
- Portable EMS in source-side shore path before camper inlet
- Single combined 6-way AC DIN enclosure with isolated AC-in and AC-out neutral paths plus common equipment grounding/PE handling
- AC-out `30A` main breaker plus `20A` / `20A` branch breaker and GFCI receptacle hardware
- Receptacle boxes + `120V` outlets (current purchased baseline `2` GFCI receptacles/covers in row `15`; inactive row `112` records the closed separate box/faceplate allowance)
- AC-out-2 reserve-only capped route (no energized Phase 1 branch hardware)
- USB PD station branch hardware (`3` stations / independent routes owner-reported `2026-08-05`: `1x` office and `2x` Galley; row `115` owns the exact purchased Acegoo evidence while the additional station purchase provenance remains untracked)
- Camper audio hardware: Kicker `46KMC2` source branch, Kicker `49PTRTP10` powered sub branch, `AUDIO-HU` 15A source protection, `AUDIO-SUB` 40A source fuse, RCA/speaker/remote wiring
- Battery temperature sensor wiring to inverter/monitoring path
- SmartShunt fused positive sense/power lead (factory harness)
- Cerbo GX `CERBO-PWR` small inline fused `48V` power feed
- Ford `Upfitter #3` control lead and local `F-15` inline fuse for the WS500 brown ignition/enable wire

Assumptions (Explicit)

1. Cable sizing assumes fine-strand copper conductors (OFC welding-cable baseline for high-current DC paths), enclosed vehicle routing, and the estimated one-way lengths listed in this document.
2. Voltage-drop design intent used here: `<=2%` on major `48V` power runs and `<=3%` on `12V` branch circuits.
3. `F-09` PV string fuse value (`15A`) remains provisional until final module datasheet max-series-fuse rating is confirmed.
4. Cerbo GX feed is now a small inline fused `48V` feed (`CERBO-PWR`) from the system/load side of the main disconnect during bench commissioning, so the Cerbo powers down with the house system.
5. Orion branch uses one source-side fuse only: `F-05 40A` MEGA, body-marked at least `58VDC`, in Lynx Slot 4 feeding the existing `6 AWG` input pair directly. The `58VDC` minimum is tied to the locked `56.8V` charge ceiling; use Victron `CIP138040020 40A/80V` if replacement is needed. Standalone `F-06` is retired; do not add an inline fuse after the Lynx slot.
6. No low-voltage-disconnect (LVD) automation is included in Phase 1; protection is source fusing plus manual `SW-12V-BATT` isolation.

7. Alternator architecture lock is dedicated `48V` secondary alternator path (`Mechman + WS500 + APM-48`) with `F-04 200A/80V`; obsolete pre-Mechman engine-bay fuse paths are removed from active layout.
8. `F-01A/B/C` are provisionally set to `200A` pending final `51.2V` battery datasheet/manual confirmation; if validated limits are lower, shift to `175A`.
9. `2/0` cable quantity planning baseline in this pass is `77.5 ft` total no-padding (`42.5 ft` red + `35.0 ft` black); user-applied order padding is intentionally deferred to checkout.
10. Manual alternator shutdown baseline is Ford `Upfitter #3` feeding the WS500 brown ignition/enable wire through `F-15`; `WS500` white `Feature-In` remains a future-only reserve for automatic interlock work.
11. Camper audio is a `12V` accessory load. KMC2 source branch is `15A`; PTRTP10 sub branch is `40A` with `4 AWG` power and return. Audio peaks can exceed Orion continuous output and rely on the `12V` buffer battery, so first loud testing should watch `12V` voltage/SOC.

Completion Status

- DC/PV topology is complete for current BOM scope and load model scope.
- AC hierarchy is complete at architecture level, including transfer behavior, branch strategy, and protection chain.
- Full-circuit estimate pass is now documented with run lengths, voltage-drop screening, and purchase rollups.
- Camper audio branch added as a DC-first `12V` accessory package: Kicker `46KMC2` source branch plus Kicker `49PTRTP10` powered sub branch; detailed product/routing notes live in `docs/implementation/CAMPER_audio_system.md`.
- Remaining work is final measured-length replacement and SKU-level closeout before final cut-to-length harness production.